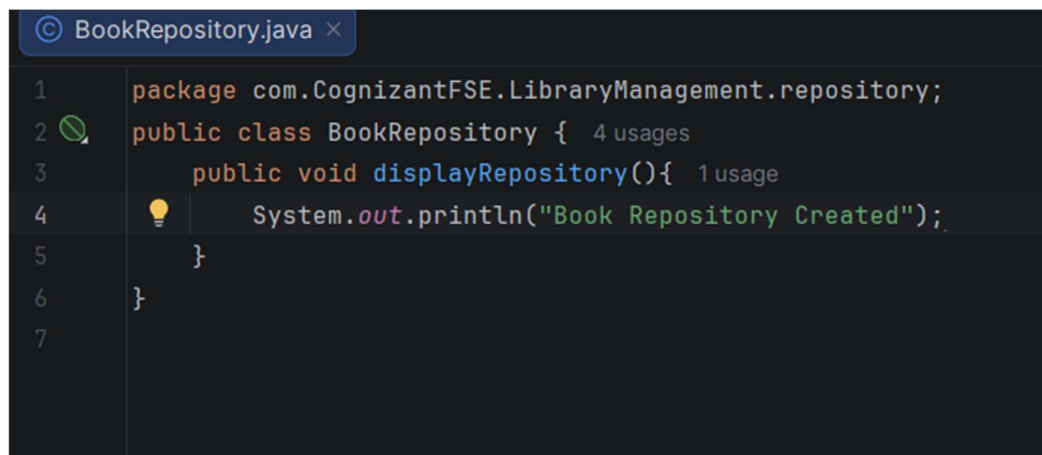
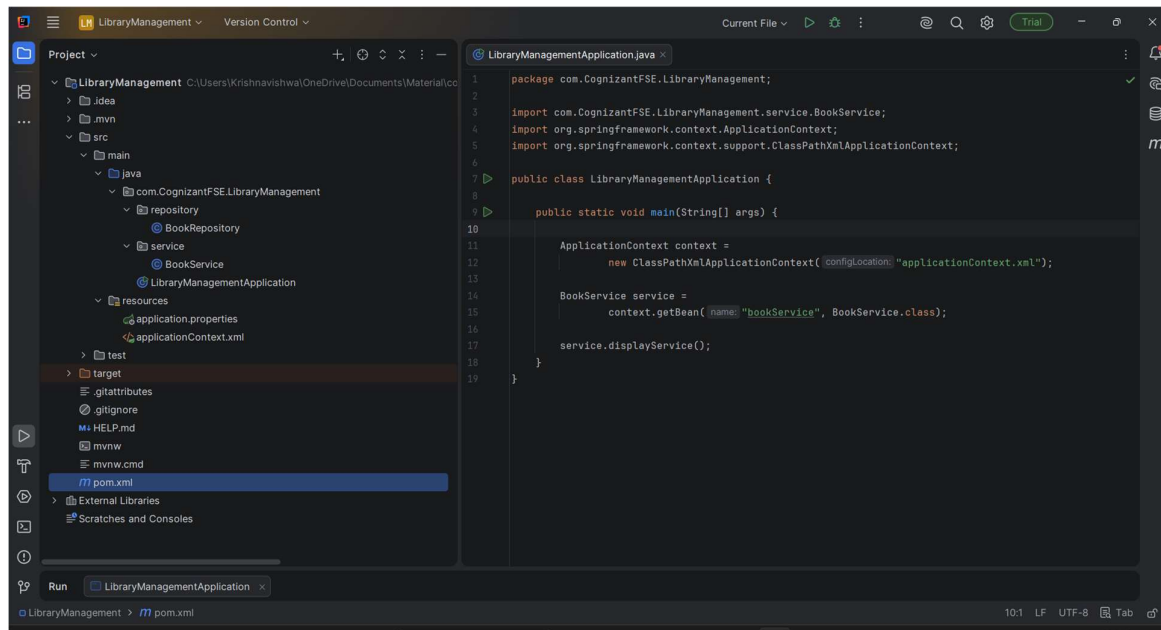


Exercise 1: Configuring a Basic Spring Application/ Exercise 2: Implementing Dependency Injection

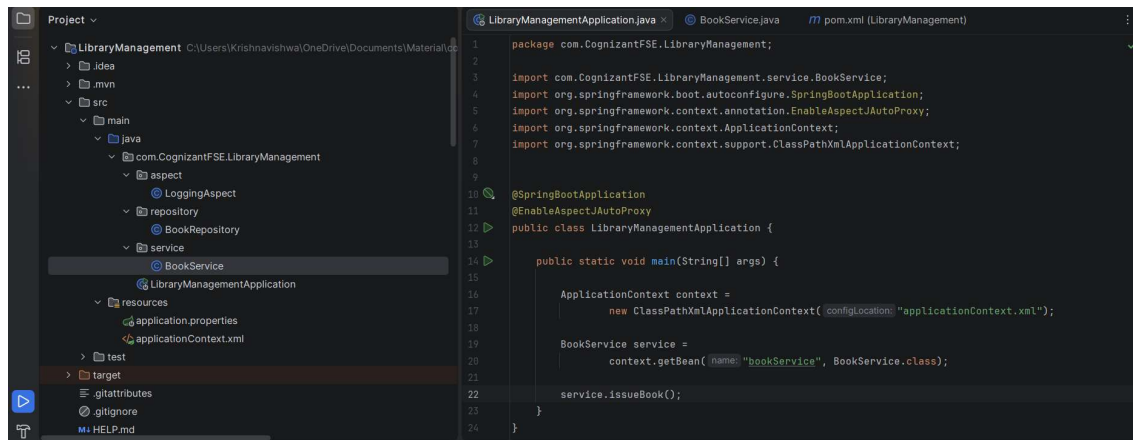


```
BookService.java x
1 package com.CognizantFSE.LibraryManagement.service;
2
3 import com.CognizantFSE.LibraryManagement.repository.BookRepository;
4
5 public class BookService { 4 usages
6     private BookRepository repo; 2 usages
7     public void setRepository(BookRepository repo){
8         this.repo = repo;
9     }
10    public void displayService(){ 1 usage
11        System.out.println("Book Service Created");
12        repo.displayRepository();
13    }
14 }
15 }
```

OUTPUT:

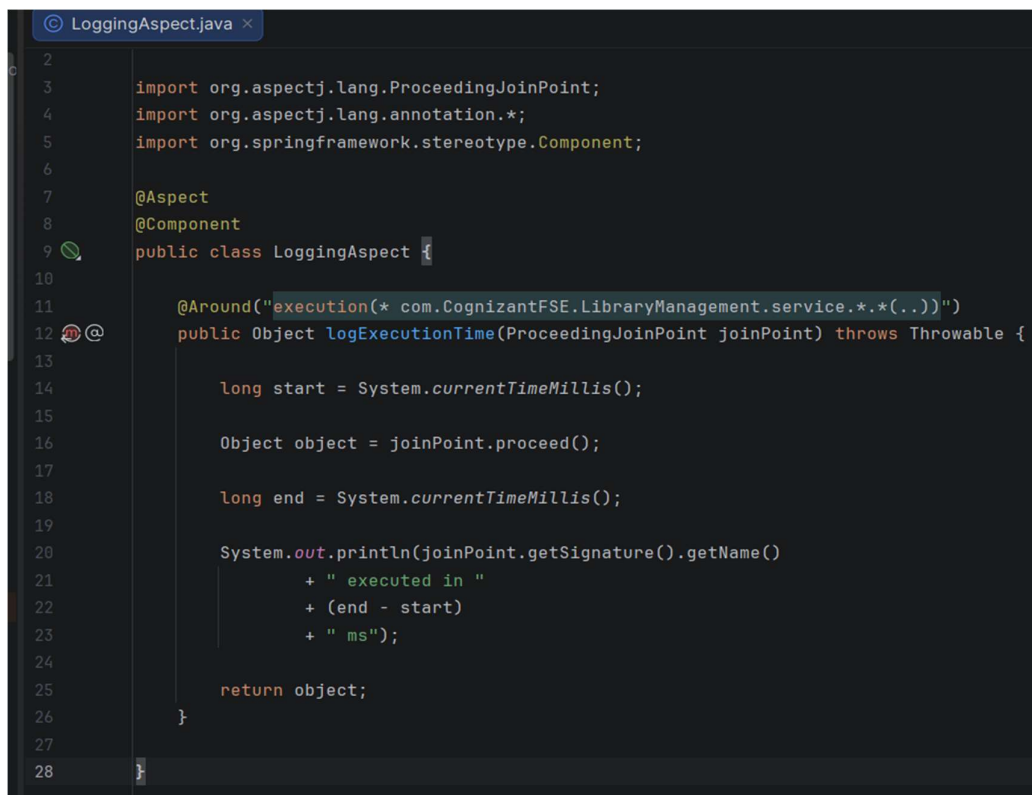
```
Run LibraryManagementApplication x
[C:\Program Files\Java\jdk-17\bin\java.exe" ...
Book Service Created
Book Repository Created
Process finished with exit code 0
```

Exercise 3: Implementing Logging with Spring AOP



The screenshot shows an IDE with a project named 'LibraryManagement'. The project structure on the left includes a 'main' directory with 'com.CognizantFSE.LibraryManagement' package, which contains 'LoggingAspect', 'BookRepository', 'BookService', and 'LibraryManagementApplication'. The 'LibraryManagementApplication.java' file is open in the editor, showing the following code:

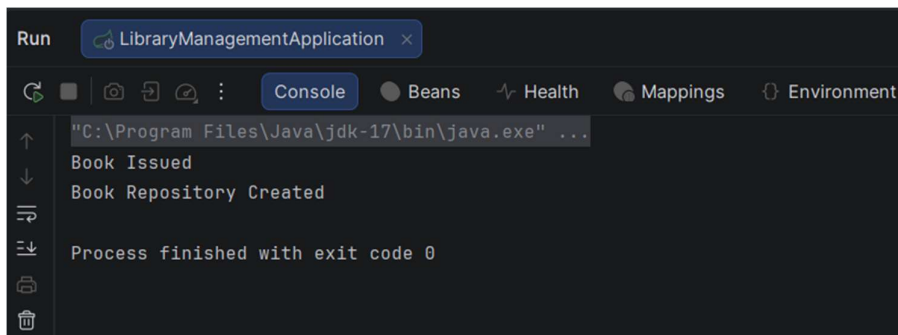
```
1 package com.CognizantFSE.LibraryManagement;
2
3 import com.CognizantFSE.LibraryManagement.service.BookService;
4 import org.springframework.boot.autoconfigure.SpringBootApplication;
5 import org.springframework.context.annotation.EnableAspectJAutoProxy;
6 import org.springframework.context.ApplicationContext;
7 import org.springframework.context.support.ClassPathXmlApplicationContext;
8
9
10 @SpringBootApplication
11 @EnableAspectJAutoProxy
12 public class LibraryManagementApplication {
13
14     public static void main(String[] args) {
15
16         ApplicationContext context =
17             new ClassPathXmlApplicationContext("applicationContext.xml");
18
19         BookService service =
20             context.getBean("bookService", BookService.class);
21
22         service.issueBook();
23     }
24 }
```



The screenshot shows the 'LoggingAspect.java' file in the IDE. The code is as follows:

```
1
2
3 import org.aspectj.lang.ProceedingJoinPoint;
4 import org.aspectj.lang.annotation.*;
5 import org.springframework.stereotype.Component;
6
7 @Aspect
8 @Component
9 public class LoggingAspect {
10
11     @Around("execution(* com.CognizantFSE.LibraryManagement.service.*.*(..))")
12     public Object logExecutionTime(ProceedingJoinPoint joinPoint) throws Throwable {
13
14         long start = System.currentTimeMillis();
15
16         Object object = joinPoint.proceed();
17
18         long end = System.currentTimeMillis();
19
20         System.out.println(joinPoint.getSignature().getName()
21             + " executed in "
22             + (end - start)
23             + " ms");
24
25         return object;
26     }
27
28 }
```

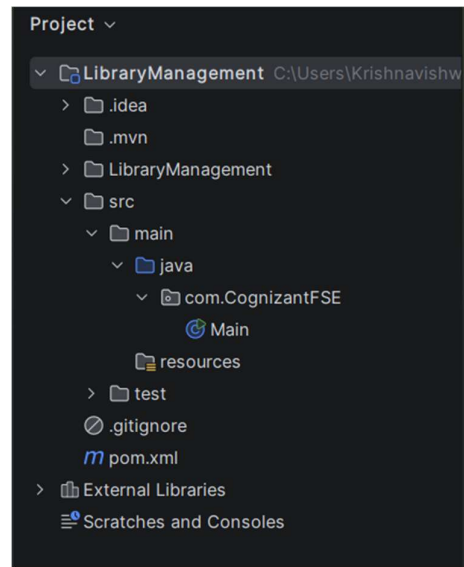
Output:



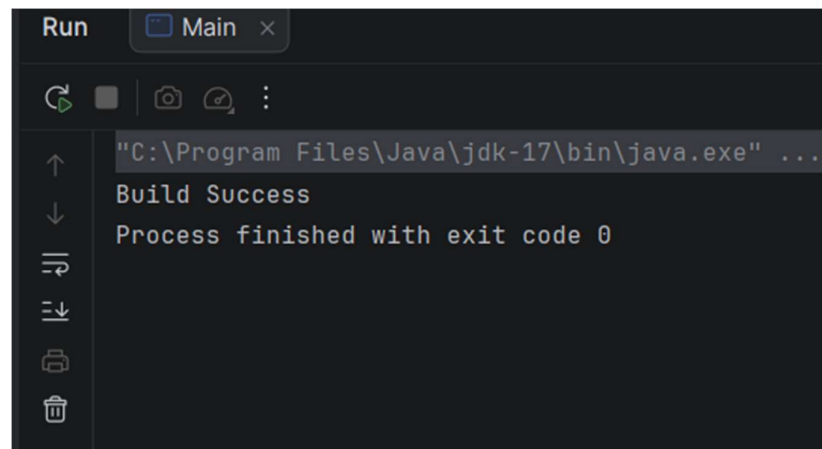
The screenshot shows the 'Run' console output for the 'LibraryManagementApplication'. The output is as follows:

```
Run LibraryManagementApplication x
C:\Program Files\Java\jdk-17\bin\java.exe ...
Book Issued
Book Repository Created
Process finished with exit code 0
```

Exercise 4: Creating and Configuring a Maven Project



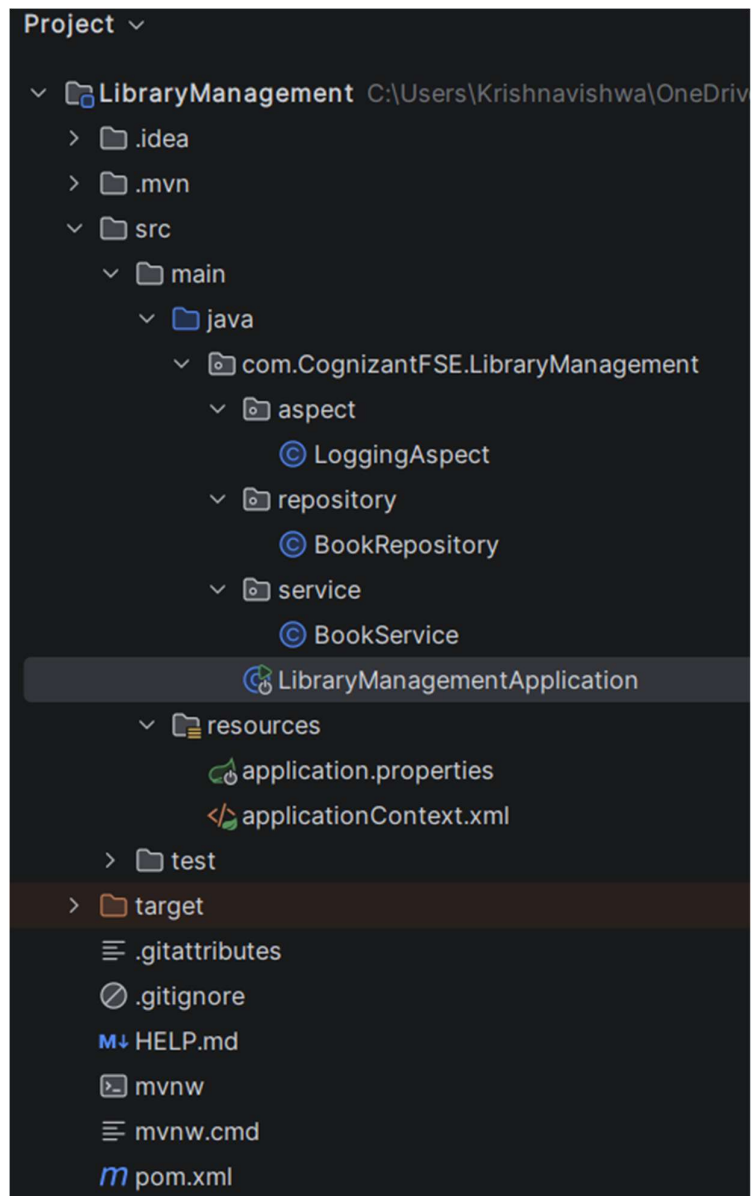
```
1 <?xml version="1.0" encoding="UTF-8"?>
2
3 <project xmlns="http://maven.apache.org/POM/4.0.0"
4         xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
5         xsi:schemaLocation="http://maven.apache.org/POM/4.0.0
6         http://maven.apache.org/xsd/maven-4.0.0.xsd">
7
8     <modelVersion>4.0.0</modelVersion> Compatible with Maven 3
9
10    <groupId>com.CognizantFSE</groupId>
11    <artifactId>LibraryManagement</artifactId>
12    <version>1.0-SNAPSHOT</version>
13
14    <properties>
15        <maven.compiler.source>17</maven.compiler.source>
16        <maven.compiler.target>17</maven.compiler.target>
17        <project.build.sourceEncoding>UTF-8</project.build.sourceEncoding>
18    </properties>
19
20    <dependencies>
21        <dependency>
22            <groupId>org.springframework</groupId>
23            <artifactId>spring-context</artifactId>
24            <version>6.2.8</version>
25        </dependency>
26
27        <dependency>
28            <groupId>org.springframework</groupId>
29            <artifactId>spring-aop</artifactId>
30            <version>6.2.8</version>
31        </dependency>
32    </dependencies>
```



The image shows a screenshot of an IDE's Run console. At the top, there is a tab labeled "Main" with a close button. Below the tab, there is a toolbar with icons for running (a green play button), a stop button, a camera icon, a speech bubble icon, and a vertical ellipsis. The main area of the console displays the following text: `"C:\Program Files\Java\jdk-17\bin\java.exe" ...`, `Build Success`, and `Process finished with exit code 0`. On the left side of the console, there is a vertical toolbar with icons for scrolling (up and down arrows), a refresh icon, a search icon, a print icon, and a trash icon.

```
Run Main x
[C] [X] [C] [S] [⋮]
↑
↓
↺
↻
🖨
🗑
"C:\Program Files\Java\jdk-17\bin\java.exe" ...
Build Success
Process finished with exit code 0
```

Exercise 5: Configuring the Spring IoC Container



ApplicationContext.xml:

```
<?xml version="1.0" encoding="UTF-8"?>
```

```
<beans xmlns="http://www.springframework.org/schema/beans"
       xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
       xsi:schemaLocation="
           http://www.springframework.org/schema/beans
           https://www.springframework.org/schema/beans/spring-beans.xsd">
```

```

<bean id="bookRepository"
      class="com.CognizantFSE.LibraryManagement.repository.BookRepository"/>

<bean id="bookService"
      class="com.CognizantFSE.LibraryManagement.service.BookService">

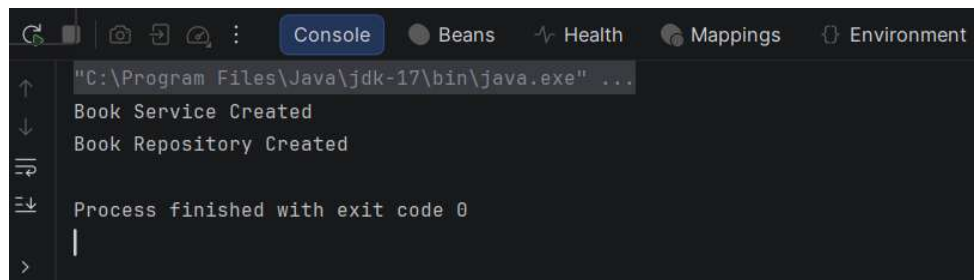
    <property name="repository" ref="bookRepository"/>

</bean>

</beans>

```

output:



```

C:\Program Files\Java\jdk-17\bin\java.exe' ...
Book Service Created
Book Repository Created
Process finished with exit code 0

```

Exercise 6: Configuring Beans with Annotations

ApplicationConfiguration.xml:

```

<?xml version="1.0" encoding="UTF-8"?>

<beans xmlns="http://www.springframework.org/schema/beans"
       xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
       xmlns:context="http://www.springframework.org/schema/context"

       xsi:schemaLocation="
http://www.springframework.org/schema/beans
https://www.springframework.org/schema/beans/spring-beans.xsd
http://www.springframework.org/schema/context
https://www.springframework.org/schema/context/spring-context.xsd">

    <context:component-scan
        base-package="com.CognizantFSE.LibraryManagement"/>

</beans>

```

```
BookService.java x  LoggingAspect.java  LibraryManagementApplication.java

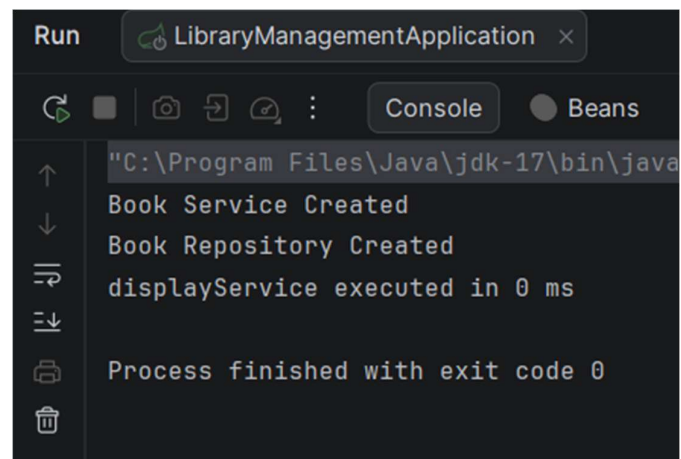
1  package com.CognizantFSE.LibraryManagement.service;
2
3  import com.CognizantFSE.LibraryManagement.repository.BookRepository;
4  import org.springframework.beans.factory.annotation.Autowired;
5  import org.springframework.stereotype.Service;
6
7  @Service 4 usages
8  public class BookService {
9
10     @Autowired
11     private BookRepository repo;
12     public void setRepository(BookRepository repo){ no usages
13         this.repo = repo;
14     }
15     public void displayService(){ 1 usage
16         System.out.println("Book Service Created");
17         repo.displayRepository();
18     }
19     public void issueBook() { no usages
20
21         System.out.println("Book Issued");
22         repo.displayRepository();
23     }
24 }
```

Here **@Service** and **@Autowired** → Annotation's

```
1  package com.CognizantFSE.LibraryManagement.repository;
2
3  import org.springframework.stereotype.Repository;
4
5  @Repository
6  public class BookRepository {
7      public void displayRepository(){ 2 usages
8          System.out.println("Book Repository Created");
9      }
10 }
```

Here **@Repository** → Annotation

Output:



The screenshot shows the 'Run' console of an IDE. At the top, there is a tab labeled 'LibraryManagementApplication' with a close button. Below the tab is a toolbar with icons for running, debugging, and other actions, along with buttons for 'Console' and 'Beans'. The console output is as follows:

```
"C:\Program Files\Java\jdk-17\bin\java  
Book Service Created  
Book Repository Created  
displayService executed in 0 ms  
Process finished with exit code 0
```

On the left side of the console, there is a vertical toolbar with icons for navigating through the output (up, down, first, last, search, etc.).

Exercise 7: Implementing Constructor and Setter Injection

```
1 package com.CognizantFSE.LibraryManagement.service;
2
3 import com.CognizantFSE.LibraryManagement.repository.BookRepository;
4 import org.springframework.beans.factory.annotation.Autowired;
5 import org.springframework.stereotype.Service;
6
7 @Service 3 usages
8 public class BookService {
9
10     @Autowired
11     private BookRepository repo;
12     Change signature
13     public BookService(BookRepository repo){
14         this.repo = repo;
15         System.out.println("Constructor Injection is Executed.");
16     }
17
18     public void setRepo(BookRepository repo) { no usages
19         this.repo = repo;
20         System.out.println("Setter Injection Executed");
21     }
22
23     public void issueBook() { 1 usage
24
25         System.out.println("Book Issued");
26         repo.saveBook();
27     }
28 }
```

Output:

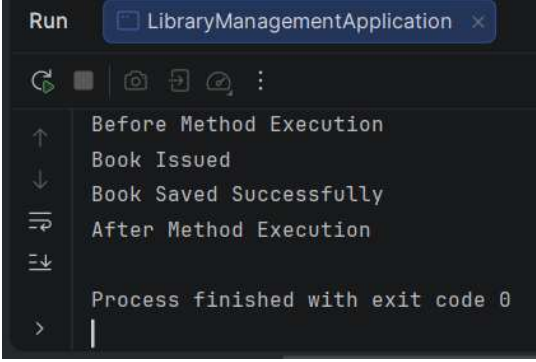
```
Run LibraryManagementApplication x
Console Beans Health
"C:\Program Files\Java\jdk-17\bin\java.exe" ...
Book Repository Created
Constructor Injection is Executed.
Book Issued
Book Saved Successfully
issueBook executed in 0 ms
Process finished with exit code 0
```

Exercise 8: Implementing Basic AOP with Spring

```
1 package com.CognizantFSE.LibraryManagement.aspect;
2
3 import org.aspectj.lang.annotation.After;
4 import org.aspectj.lang.annotation.Aspect;
5 import org.aspectj.lang.annotation.Before;
6
7 @Aspect
8 public class LoggingAspect {
9
10     @Before("execution(* com.CognizantFSE.LibraryManagement.service.BookService.issueBook(..))")
11     public void beforeMethod() {
12         System.out.println("Before Method Execution");
13     }
14
15     @After("execution(* com.CognizantFSE.LibraryManagement.service.BookService.issueBook(..))")
16     public void afterMethod() {
17         System.out.println("After Method Execution");
18     }
19 }
20
```

```
1 package com.CognizantFSE.LibraryManagement;
2
3 import org.springframework.context.ApplicationContext;
4 import org.springframework.context.support.ClassPathXmlApplicationContext;
5
6 import com.CognizantFSE.LibraryManagement.service.BookService;
7
8 public class LibraryManagementApplication {
9
10     public static void main(String[] args) {
11
12         ApplicationContext context =
13             new ClassPathXmlApplicationContext("applicationContext.xml");
14
15         BookService service =
16             context.getBean("bookService", BookService.class);
17
18         service.issueBook();
19     }
20 }
```

Output:



```
Run LibraryManagementApplication x
Before Method Execution
Book Issued
Book Saved Successfully
After Method Execution
Process finished with exit code 0
```